

Toward a Detector Foundation Model

Jet calibration on two merged student pipelines — what we did, what we measured, what we learned

A quick feasibility pass on opportunistic Run 3 / HL-LHC subsamples — the groundwork for a properly scaled study.

187,652

calorimeter cells bridged exactly between
the two formats

1.20M

matched jets in the calibration datasets
(620k light)

~115

trainings across six gated study campaigns

× 0.50

JER vs the calibrated-jet baseline at 65–90
GeV

Amir Farbin · Umar (5D ntuples, b-tagging) · Mohammad Ali (Calo ntuples, foundation model) — UT Arlington, August 2026

Two student pipelines, two ntuple formats

5D ntuples — Umar / SLAC production (labeled)

- Per-event tracks (16 stored vars + 2 derived significances), AntiKt4EMTopo jets with truth matching, truth record
- Thinned calorimeter cells in GeV — no cell IDs, no stable ordering
- 17 $\bar{t}t$ files used: the labeled sample for every supervised study

Calo ntuples — Mohammad Ali / UTA production (unlabeled fuel)

- All 187,652 cells per event (MeV, hash-ordered) + topoclusters
- Precomputed cell neighbor graph — the geometry asset we transfer
- Corpora: $\bar{t}t$ (1k events), single- π (5k), $hh \rightarrow b\bar{b}\tau\tau$ (10k) for self-supervised pretraining

Feasibility framing — read every result with this in mind

These are subsamples we happened to have, from different eras (Run 3 / HL-LHC) and different productions. Generator, tune and pile-up provenance is still being collected from the producers. All comparisons are internally consistent — same jets, same estimators, same splits — but nothing here is a tuned-performance claim.

The bridge: hacks that make two formats one dataset

- 1 **Exact geometry match** — per-layer KD-tree on cell (x, y, z) + sampling; coordinates are bit-identical across formats, recovering the hash index. Zero mismatches in 146k jets tested.
- 2 **Persistent Cell-ID map** — geometry resolves each new ID once; afterwards an O(1) lookup.
- 3 **Units & ordering** — MeV → GeV; hash order → compact indices of the processed cell corpus.
- 4 **Graph transfer** — the Calo neighbor graph, restricted to each jet's (or event's) matched cells by subgraph extraction — 5D cells get a graph they never had.
- 5 **Noise guard** — single-cell $|E| > 100$ GeV veto (forward pile-up noise, out-of-time undershoot) instead of event vetoes.

What the 5D thinning actually is — measured, not assumed

- Cells kept: $E > 0.5$ GeV AND $E > 2\sigma(\text{noise})$, positive cells only
- Retains 75.0% of jet cell energy — layer-dependent:
EMB1/EMB3 $\approx 33\%$, Tile $\approx 81\text{--}86\%$
- The model therefore sees a biased cell subset — documented and accepted for feasibility; only a new production fixes it

Lesson: geometry is a universal key between detector data formats.

The task: probabilistic jet-pT calibration

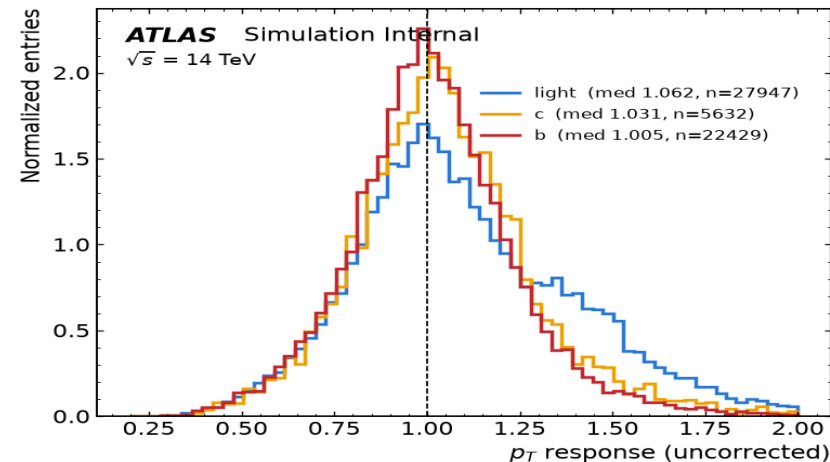
Target $y = \ln(pT_{\text{true}} / pT_{\text{reco}}) \rightarrow \text{corrected } pT = pT_{\text{reco}} \cdot e^{\mu}$

The model predicts (μ, σ) : a correction and a per-jet uncertainty.

Jet selection

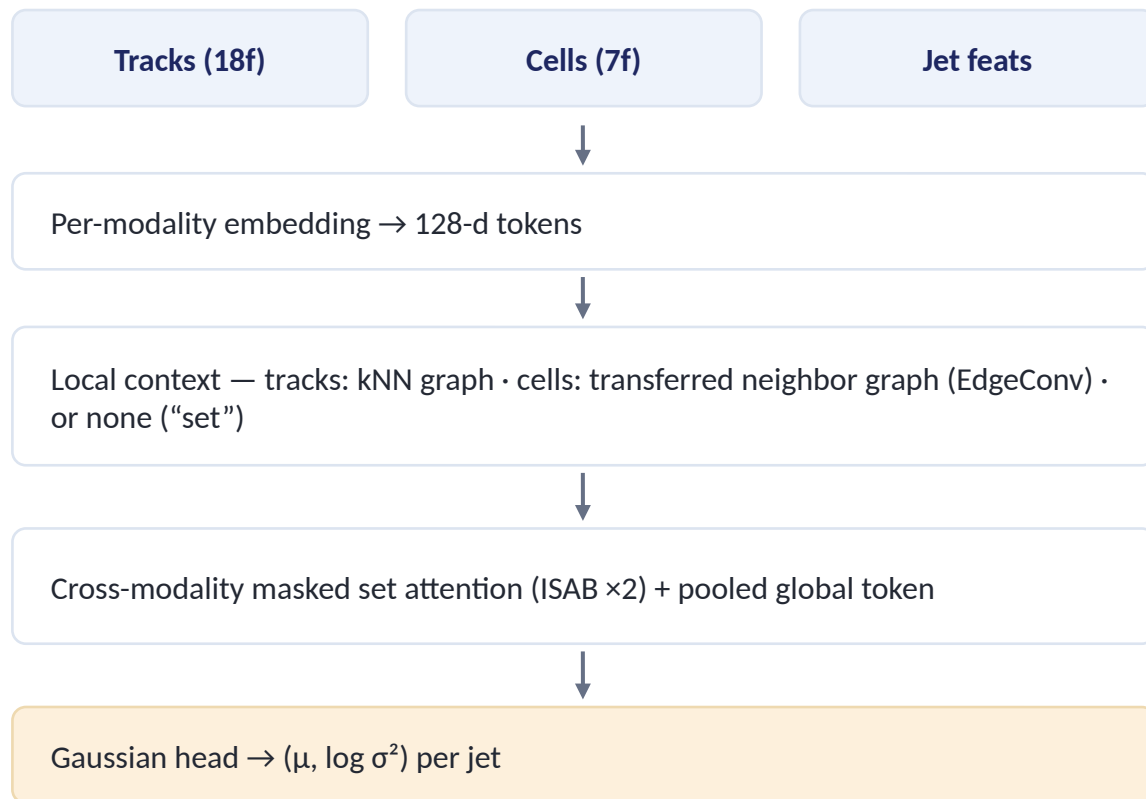
- Truth-matched AntiKt4EMTopo jets; $pT_{\text{true}} > 20 \text{ GeV}$; $|\eta| < 2.5$
- $\Delta R(\text{reco}, \text{truth}) < 0.3$ recomputed from kinematics + duplicate-match guard
- Isolation flag stored — train on isolated jets, evaluate both populations (the non-isolated gap probes inter-cluster energy sharing)
- Per jet: cone tracks · jet features · $\Delta R < 0.4$ cells + transferred graph

Item	Value
Light-jet campaign	620k jets — 181k isolated training jets
Splits (by file)	13 train / 2 validation / 2 test — test files never touched
All-flavor extension	1.20M matched jets (light : c : b mix of $t\bar{t}$)
Training weights	flat in log pT, 20–250 GeV, capped at 99th pct



Uncorrected response by flavor — the problem being fixed.

One shared encoder — the options and how we label them



≈ 1M parameters · identical backbone in every study — only the input set and the head change.

Label	Meaning
T / J	tracks in the jet cone (18 features) / jet-level features of the reco jet
C-graph / C-set	cone cells with the transferred neighbor graph / as a plain set
TJ, TC, TJC ...	input combinations (token-set concatenation)
-pre / -probe	initialized from masked-cell pretraining / frozen encoder + linear head
-f0.01 / -f0.1	trained on 1% / 10% of the labels (label-efficiency scans)
+fc	flavor-conditioned calibration (truth flavor as input)
+pan	panoptic head (p_b, p_c, p_light, μ , σ): tagging + calibration jointly
B0	baseline: the ntuple's own calibrated jets, evaluated identically

How the models are trained — and how we score them

Loss: short Huber warm-up (stability), then heteroscedastic Gaussian NLL: $L = \frac{1}{2} \ln \sigma^2 + (y - \mu)^2 / 2\sigma^2$

Optimizer: AdamW + cosine schedule; early stopping on validation NLL

Weights: flat in log pT so no momentum region dominates the fit

Seeds: 3 per configuration; mean $\pm$ spread reported (10-seed convention planned)

Hygiene: splits by file; every normalization and weight fit on the training files only

Ops: shared university GPU node — thread-capped, nice'd, watchdog-guarded; ~100 trainings without incident after hardening

Metrics (per pT-true bin; also profiled vs η , ϕ)

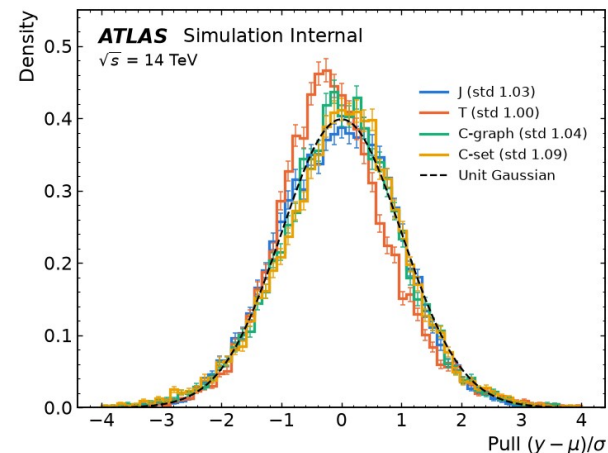
JES = median r, $r = pT_{corr} / pT_{true}$

JER = $(q75 - q25) / 1.349 \div \text{median}$ — robust, tail-insensitive

pull $z = (y - \mu) / \sigma \rightarrow \text{std} \approx 1 \Leftrightarrow$ honest uncertainties (we get 1.00–1.05)

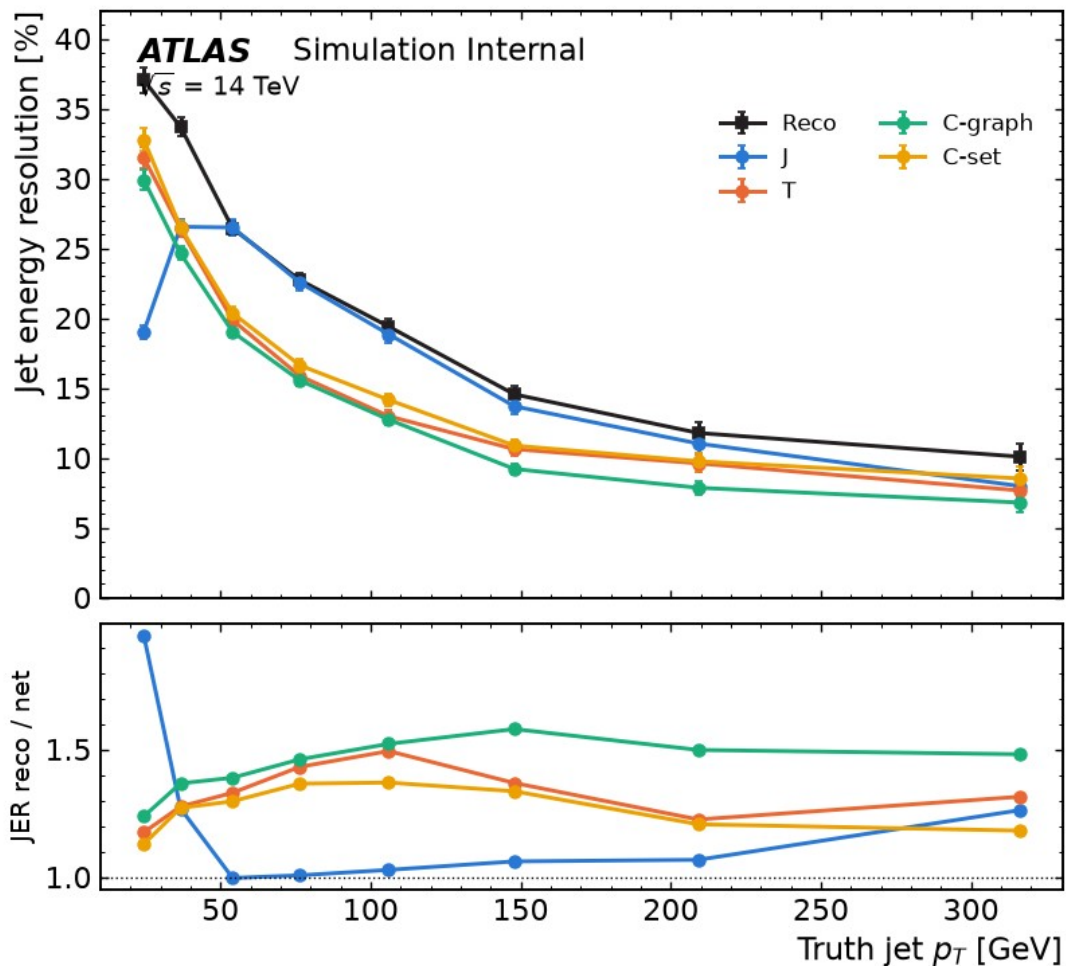
closure = RMS over bins of $(JES - 1)$

Bins: 20–400 GeV, headline 65–90 · B0 measured identically on the same test jets · details in backup



Pull distributions — predicted σ matches observed spread.

Which inputs matter? (light jets, all 17 files)



JER vs p_T , single-input models vs the stored calibrated jets (Reco); ratio below.

Model	JER @ 65–90	÷ B0
B0 (calibrated jets)	0.230	1.00
C-graph (cells only)	0.158	0.69
T (tracks only)	0.150	0.65
TC-graph	0.130	0.57
TJ	0.125	0.54
TJC-set	0.116	0.51
TJC-graph	0.115	0.50

Tracks are the backbone; cells pay off only with enough data. At 4 files cells added nothing over TJ — at 17 files they cut JER a further 8% ($0.125 \rightarrow 0.115$). Ablation conclusions are scale-dependent.

Jet features still help (TC 0.130 $\rightarrow$ TJC 0.115): “J is learned from the cells” is not yet demonstrated at this scale.

Graph vs set: a near-tie at full data (0.115 vs 0.116); the graph mattered most in cell-only and low-data regimes.

Does self-supervised pretraining transfer?

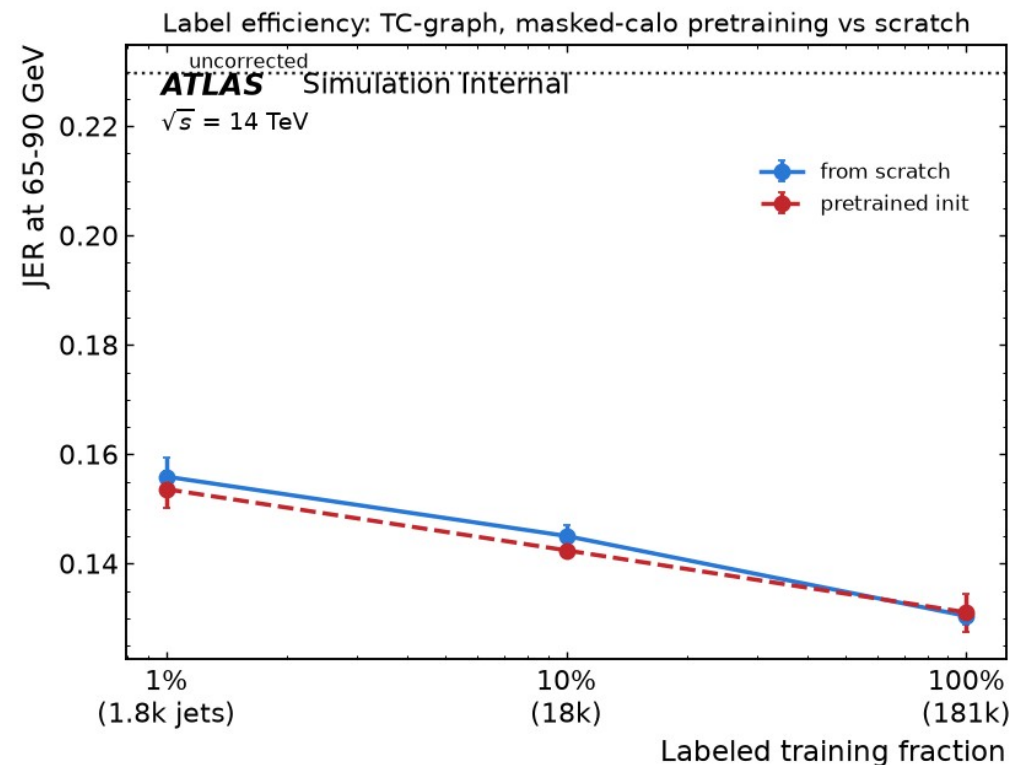
Pretext: mask groups of cell tokens on the Calo corpora and reconstruct their features; reconstruction QA validated against truth (Note 01 figure set).

Corpus scale helps: masked-reconstruction validation loss 0.507 (ft-only) $\rightarrow$ 0.347 with the 3-corpus mix ($\approx$ 14k events).

Transfer is real: a linear probe on the frozen pretrained encoder matches full end-to-end training — JER 0.157 vs 0.158 (cells-only). The representation already carries calibration-relevant structure.

Label efficiency: with 1% / 10% of labels, pretrained initialization is consistently but modestly better (0.156 $\rightarrow$ 0.154, 0.145 $\rightarrow$ 0.142).

Caveat: pretraining and fine-tuning samples differ in (unverified) provenance — matched provenance and larger corpora are the obvious next step.

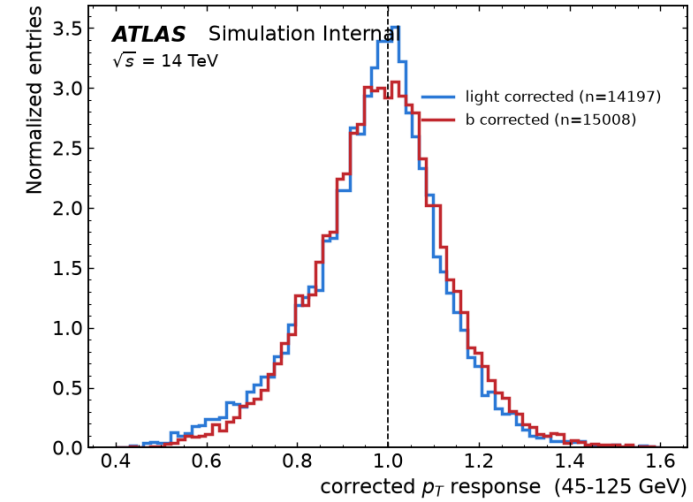


Label-efficiency scan: pretrained vs from-scratch at 1%, 10%, 100% of labels.

One model for flavors — one head for tagging + calibration

T1 — flavor-aware calibration (Note 02)

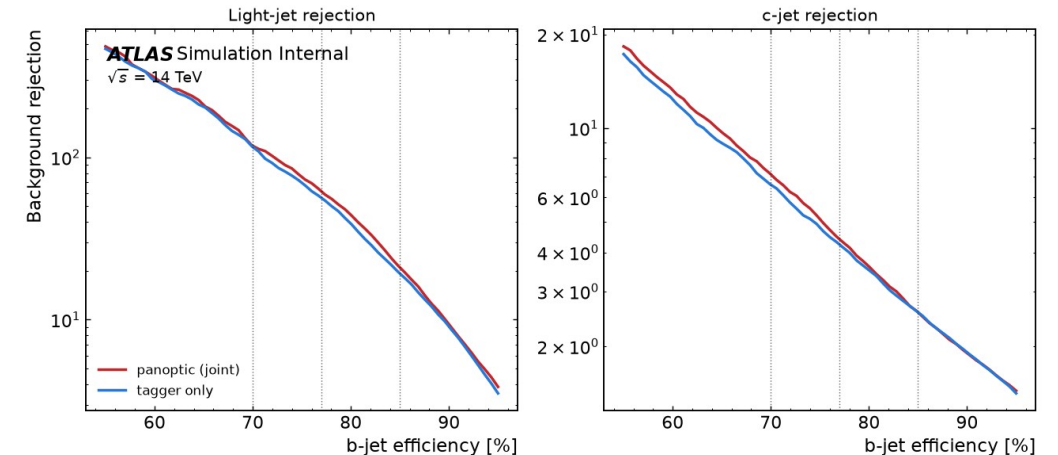
- Dedicated per-flavor models lose to one shared model — JER c: 0.154 $\rightarrow$ 0.131, b: 0.138 $\rightarrow$ 0.128: cross-flavor statistics beat specialization at 10^6 jets
- Flavor-conditioned shared model is best overall (closure 3.9%)
- b-response shoulder = semileptonic decays (physics, not a model artifact) $\rightarrow$ motivates mixture-density heads



b-jet response: the semileptonic shoulder.

T2 — panoptic reconstruction (Note 03)

- One head outputs (p_b , p_c , p_{light} , μ , σ) — tagging and calibration jointly
- AUC(b) 0.941 ± 0.001 vs 0.939 for an identical dedicated tagger; light rejection 117 ± 5 at the 70% WP; c-rejection consistently better
- Calibration within $\sim 2\%$ of the truth-conditioned optimum (JER 0.126 vs 0.123) — the model supplies its own conditioning signal



Background rejection vs b-efficiency: panoptic = dedicated tagger.

Lessons learned — and where this goes

- 1 **Geometry bridges formats exactly** — the neighbor graph transfers; two codebases now share one encoder.
- 2 **ML calibration halves JER** — 0.115 vs 0.230 at 65–90 GeV against the stored calibrated jets (indicative comparison).
- 3 **Ablations are scale-dependent** — cells looked useless at 4 files, essential at 17 — beware small-sample conclusions.
- 4 **Pretraining transfers** — a linear probe matches full training; corpus diversity helps; gains modest so far.
- 5 **Sharing beats specializing** — one model for all flavors, one head for tagging + calibration, at no cost.
- 6 **Shared-node ops need engineering** — thread caps + watchdogs; ~100 trainings without further incident.

The rest of the program

- Done: cell-selection bias study · masking-pretraining QA · Notes 01–03 in LaTeX, in the repo
- New: learned MET beats classical estimates by 37% (T3); one-shot DETR jet finding demonstrated (T4) — backup slides
- Next: close the gap to ATLAS-PUB depth — mixture-density heads, coverage/pull validation, 10 seeds, HPO, provenance (phased, gated plan in repo)

Asks

- Sample provenance metadata from both productions
- A larger + second-process track-level (5D) production
- A same-event track + cell corpus for matched pretraining

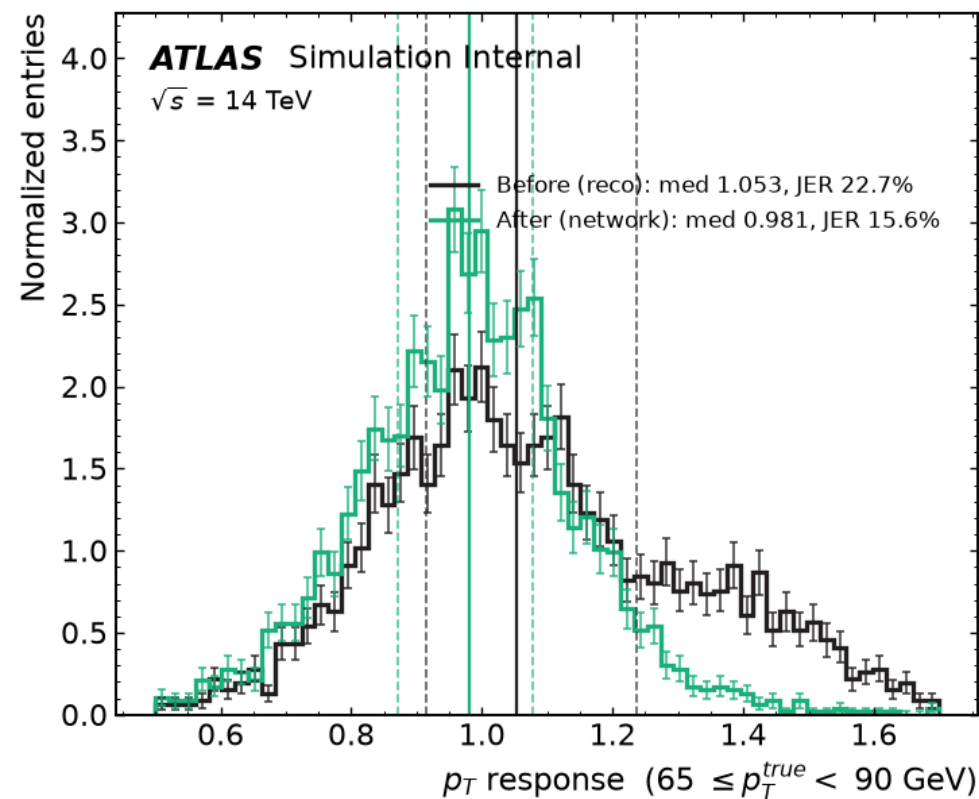
Backup

Measuring jet energy scale, resolution, and uncertainty honesty · event-level results (T3 MET, T4 jet finding)

From per-jet predictions to JES and JER

- 1 **Per jet** — the model gives (μ, σ) for $y = \ln(pT_true/pT_reco)$; corrected $pT = pT_reco \cdot e^{\mu}$; response $r = pT_corr / pT_true$.
- 2 **Bin** — in pT_true with edges [20, 30, 45, 65, 90, 125, 175, 250, 400] GeV; also vs η (crack region its own bin) and vs ϕ .
- 3 **Scale** — $JES(bin) = \text{median}(r)$ — robust location.
- 4 **Resolution** — $JER(bin) = (q75 - q25)/1.349 \div \text{median}(r)$ — Gaussian-equivalent width, made fractional by the scale.
- 5 **Uncertainties** — spread across 3 training seeds (mean $\pm$ std); 10-seed convention planned.

Why robust estimators: response tails are non-Gaussian (semileptonic b, pile-up); mean/std would be tail-dominated. A mean-seeking loss variant (MAE study) shifts closure by $\sim 2\%$ — estimator and loss must agree.



Metric construction, step by step, on the test sample.

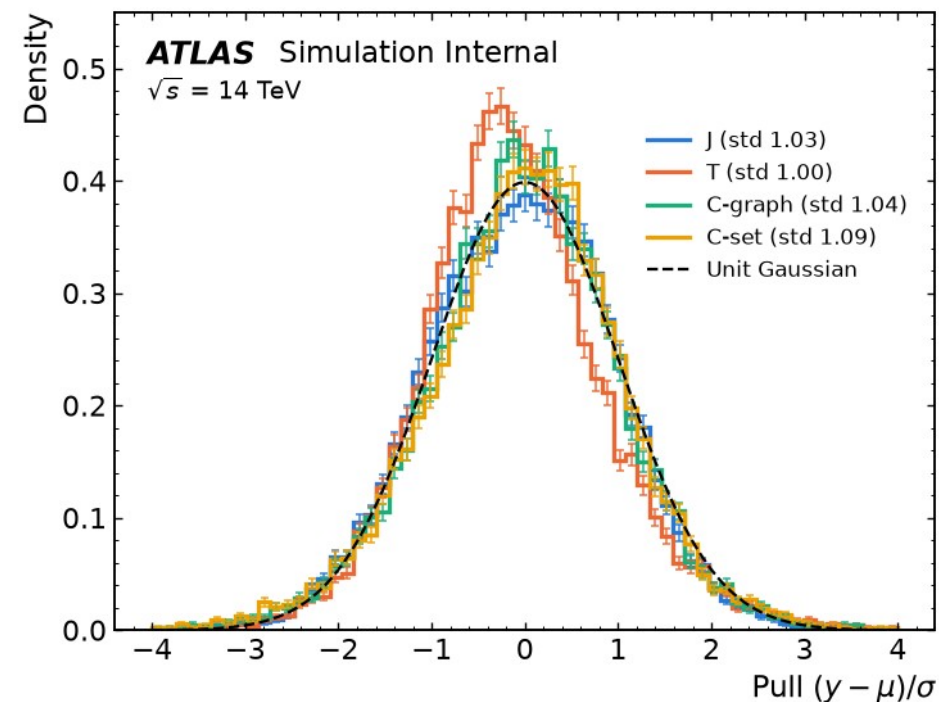
Uncertainty honesty and closure

Pull: $z = (y - \mu)/\sigma$ per jet. If the predicted σ is honest, z has unit width. Measured pull std: 1.00–1.05 across configurations.

Closure: RMS over pT bins of $(JES - 1)$. Best configurations reach 3.9–4.4%; a known low-pT non-closure (20–30 GeV) remains — numerical inversion planned.

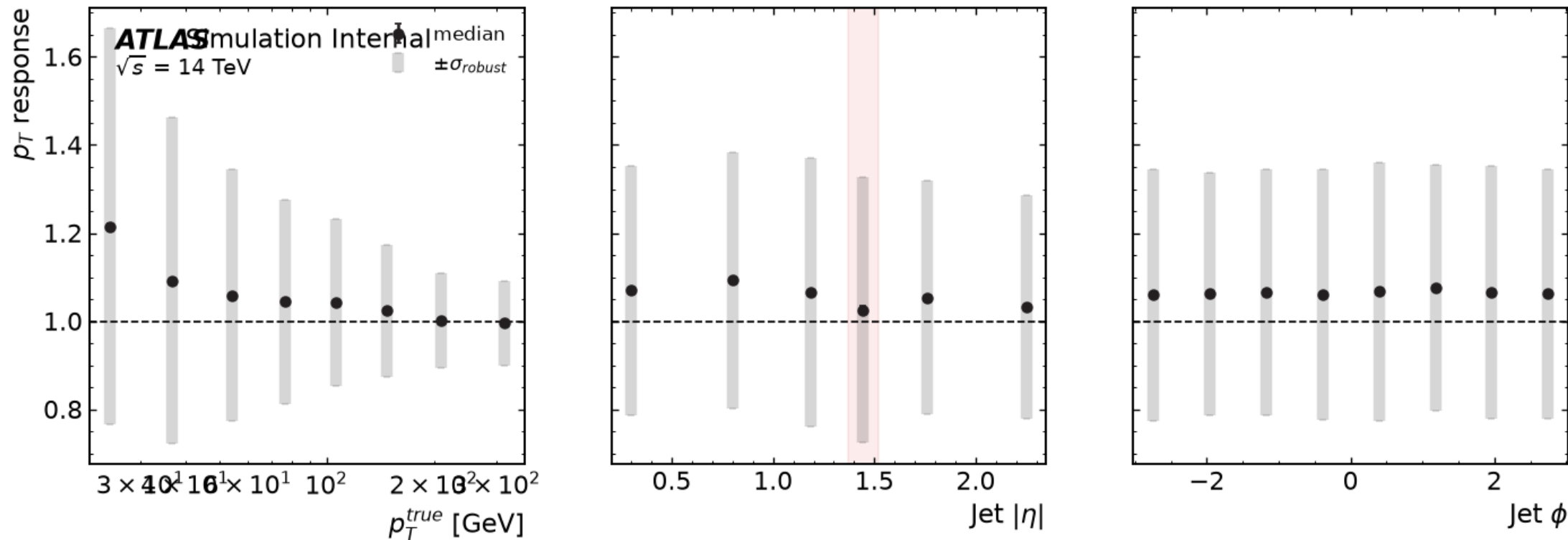
What this does not yet show: coverage diagrams, per-bin χ^2 , and tail fractions $P(|z| > k)$ — the PUB-note-grade validation suite. Scheduled in the gap-closure plan; it runs on archived predictions, no retraining needed.

Fit/apply hygiene: all normalizations, weights, and any rescaling are fit on training or validation files and applied frozen to test files.



Pull distributions vs a unit Gaussian, per configuration.

The B0 baseline and fair-comparison caveats



JES and JER profiles vs p_T , η , ϕ — model vs the stored calibrated jets (B0), identical test jets and estimators.

B0 = the calibrated jet p_T already stored in the ntuple, evaluated with the identical binning and estimators on the identical test jets.

Caveats: that calibration was derived elsewhere (different target, samples, estimator conventions); our sample is $\bar{t}t$ -only and, for the headline, light-quark isolated jets; aggregation is jet-weighted.

Therefore “JER $\times 0.50$ ” is an internally consistent, indicative comparison — not a claim against the official in-situ calibration chain.

T3: MET estimation from tracks + cells

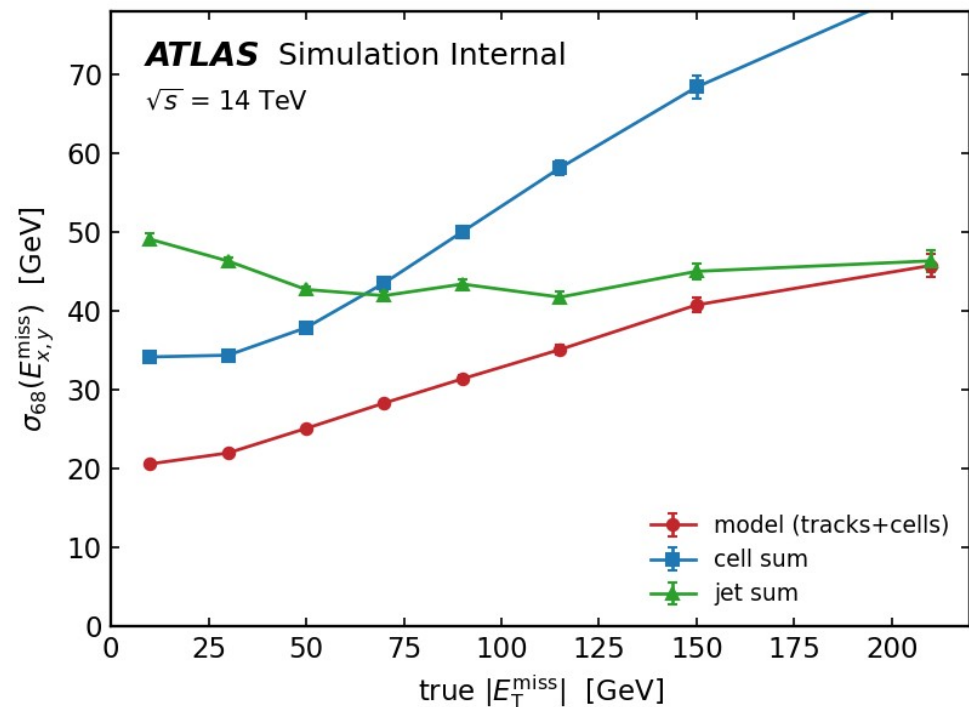
Target: truth MET = vector sum of truth-record neutrinos. One sample = one event: all PV/unassociated tracks ($p_T > 0.5$ GeV, ≤ 500) and all matched cells with the transferred event neighbor subgraph.

Head: pooled global token $\rightarrow$ (MET_x, MET_y); Huber loss on MET/100 GeV.

Baselines: negative vector sums of tracks / cells / matched jets, computed on the identical test events.

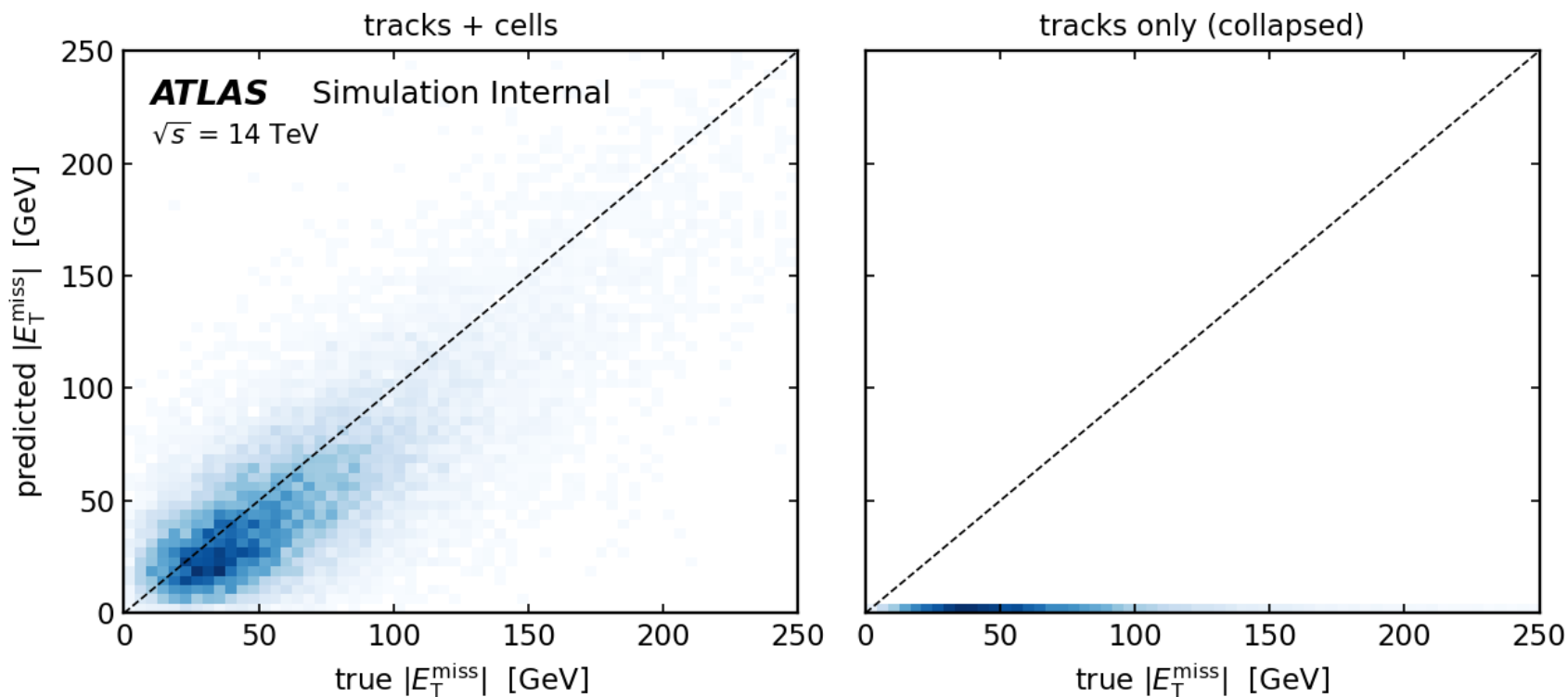
Estimator	σ_{68} per component [GeV]
track sum	60.6
cell sum	43.2
jet sum	44.2
model, tracks only	49.5 — collapsed to constant output
model, tracks + cells	27.0 ± 0.1 (3 seeds) -37% vs best classical

Improvement is uniform over the full MET range; the cell-sum baseline degrades steeply once true MET is large.



Resolution vs true $|MET|$: learned estimate vs classical vector sums, identical test events.

T3: where the MET information lives



Predicted vs true $|MET|$, test events: the same encoder with cells (left) and without (right).

Tracks alone collapse: every seed converges to a near-constant output (prediction spread ≈ 0 , correlation ≈ 0) — charged recoil carries almost no extractable neutrino information at this scale, and the optimizer finds the trivial minimum.

Cells flip the outcome: the identical encoder, given the matched cells and their transferred neighbor graph, correlates strongly with truth — MET is a calorimeter measurement, and the event-level graph the cell-matching bridge enables is doing real work.

T4: one-shot jet finding via set prediction

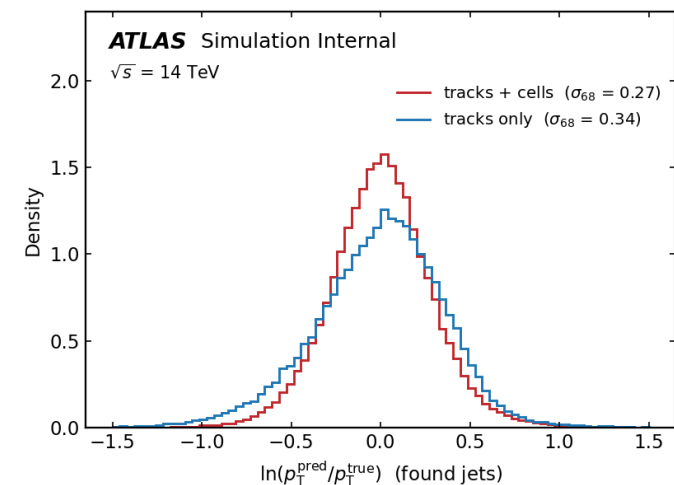
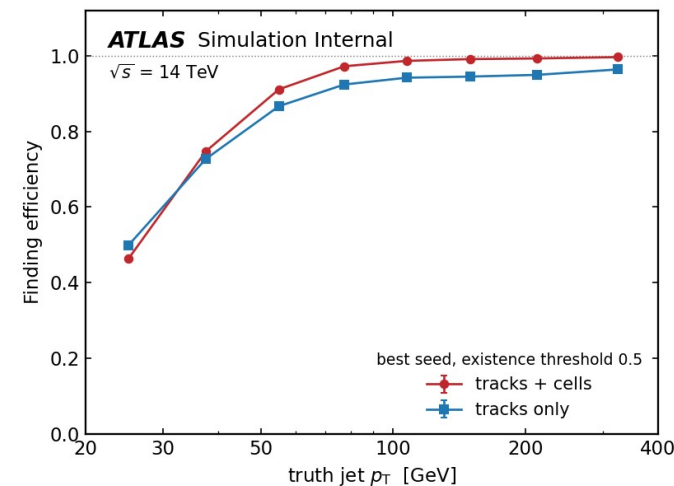
Method (DETR-style): 12 learned queries cross-attend to the event tokens; per query (existence, log pT, η , ϕ). Hungarian-matched set loss: existence BCE + kinematic L1 on matched pairs. Truth: hard-scatter jets, $p_T > 20$ GeV.

Metric (3 seeds)	tracks	tracks + cells
efficiency ($p_T > 20$)	0.789 ± 0.010	0.759 ± 0.052
fake rate	0.125 ± 0.021	0.110 ± 0.016
log-pT MAE (found)	0.293 ± 0.010	0.238 ± 0.015

Read the profile, not the integral: the scalar efficiency is dominated by the 20–30 GeV turn-on; at plateau (> 80 GeV) the best seed reaches 0.99 with cells vs 0.95 without, and cells sharpen the found-jet kinematics.

DETR instability is real: large seed variance with cells (best-seed val 1.31 vs 1.71) — matching-stabilization techniques are the known fix and the obvious next step.

Rerun note: TC seeds trained under torch 2.13 after the reboot-forced venv rebuild; no checkpoints cross versions.



Top: finding efficiency vs truth p_T (best seed). Bottom: log- p_T residual of found jets.